The Adrian–Ciel Law: Numerical Resonance of the Divisor Ratio, Linear Complexity Charge $\Lambda^0(\omega)$, and Phase Structure in the Odd Integers

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Abstract

We report a computational study of the divisor ratio $R(n) = \sigma(n)/n$ for $n \leq 3 \times 10^6$, with emphasis on odd integers. We discover robust resonance bands at $R \approx 1.20$ and $R \approx 1.37$ that correlate with the number of distinct prime factors $\omega(n)$. We introduce an informational complexity charge $\Lambda^0(\omega)$ via per-ω medians of $\log R$, empirically linear: $\Lambda^0(\omega) = a + b\,\omega$ with $R^2 \approx 0.99$. We formulate a Lagrangian $\mathcal{L}(n)$ combining the baseline Λ^0, a harmonic divisor term, and a quadratic perfection penalty, and we state the Adrian–Ciel Law describing the interplay of ω, parity, and resonance bands. This manuscript is written as a methodical research notebook: step-by-step methodology, iterated conclusions, and ready-to-fill figure/table placeholders enabling reproduction.

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1 Introduction

Problem statement. Let $\sigma(n)$ denote the sum-of-divisors function and define the *divisor ratio*

$$R(n) \;=\; \frac{\sigma(n)}{n}. \tag{1}$$

Perfect numbers satisfy $R(n) = 2$, while abundant/deficient classes are characterized by $R(n) \gtrless 2$. Instead of searching directly for perfect numbers, we study the *statistical structure* of $R(n)$ across large ranges of n, asking whether R exhibits stable *resonances* (density bands) and how these relate to arithmetic complexity.

Guiding hypothesis. There exists a low-dimensional parameter controlling R's baseline behavior, with additional discrete "phase-like" structures superimposed. The natural candidate is the number of distinct prime factors,

$$\omega(n) = \#\{p : p \text{ prime},\ p \mid n\}.$$

We therefore analyse R *conditioned on* ω.

Contributions.

1. Identification of stable resonance bands in $R(n)$ for odd integers, notably near $R \approx 1.20$ and $R \approx 1.37$.
2. Definition and empirical linear law for the *informational complexity charge*

$$\Lambda^0(\omega) := \operatorname{median}\Big(\log R(n) \mid \omega(n) = \omega\Big) = a + b\,\omega \quad (R^2 \approx 0.99).$$

3. Phase classification (BIF, CRYS, MANI) and a Lagrangian $\mathcal{L}(n)$ balancing baseline, harmonic content, and distance to perfection $R = 2$.
4. A *parity effect*: for even n, powers of two act as a raising operator on modal R at fixed ω, smearing odd-class bands toward $R \simeq 2$.

Interim conclusion. *The divisor ratio $R(n)$ over large ranges is not featureless. Conditioning on $\omega(n)$ exposes a linear baseline in $\log R$ and reveals discrete resonance bands for odd integers, suggesting a phase-like arithmetic phenomenon.*

2 Definitions and Notation

Core quantities. Throughout this study we use the following definitions:

- $\sigma(n) = \sum_{d \mid n} d$ — sum-of-divisors function.
- $R(n) = \dfrac{\sigma(n)}{n}$ — divisor ratio (Eq. 1).
- $\omega(n) = \#\{\, p : p \text{ prime}, p \mid n \,\}$ — number of distinct prime divisors.

Informational charge $\Lambda^0(\omega)$. For a fixed number of distinct prime factors ω, define

$$\Lambda^0(\omega) = \text{median}\Big(\log R(n) \,\Big|\, \omega(n) = \omega \Big), \tag{2}$$

empirically approximated by the sample median of $\log R$ for all integers with given ω. It serves as the baseline or "solo" term of informational complexity.

Phase bands. Empirical analysis revealed three main R–bands, which we treat as *phases*:

Label	**Range in $R(n)$**	**Interpretation**
BIF	$R \in [1.00, 1.10]$	Bifurcation / informational onset
CRYS	$R \in [1.18, 1.24]$	Crystallization / symbolic coherence
MANI	$R \in [1.34, 1.40]$	Manifestation / $\alpha-$resonance

Parity notation. Odd and even classes are denoted by subscripts:

$$R_{\text{odd}}(n), \qquad R_{\text{even}}(n),$$

and will later be compared in §9 to isolate the parity effect.

> ***Interim conclusion.*** *These definitions supply the algebraic backbone for our later empirical laws. The parameter ω quantifies structural complexity, while R and its bands capture emergent "phase" behaviour in divisor geometry.*

3 Experimental Setup

Scope. We analysed all integers $n \leq 3 \times 10^6$, restricting to odd numbers for primary resonance detection, and comparing with even numbers for validation.

Computational pipeline. The workflow consisted of three sieve-based stages:

1. **Sigma sieve.** For each $i \in [1, N]$, increment all multiples $k = i, 2i, 3i, \ldots$ by i:

   $$\sigma(k) \mathrel{+}= i.$$

   This additive sieve computes $\sigma(n)$ for all $n \leq N$ in $O(N \log N)$.

2. **Prime sieve and ω accumulation.** Mark primes via Eratosthenes; for each prime p, increment $\omega(k)$ for all multiples k of p.

3. **Derived quantities.** Compute $R(n) = \sigma(n)/n$, $\log R(n)$, and phase membership via the thresholds above.

Verification and controls. Cross-checked $\sigma(n)$ against known perfect numbers up to 10^5 and verified that $R(n) = 2$ at $n = 6, 28, 496, 8128$. Sampled random n for consistency between σ and explicit factorization.

Table 1: Number of integers by distinct prime factors (ω) and parity class. Column Λ_0^{emp} shows the median $\log R(n)$ conditioned on $\omega(n) = \omega$.

ω	Odd Count	Even Count	Median $\log R$ (odd)	Median $\log R$ (even)
1	217 150	21	0.000 001	0.692 903
2	567 860	243 970	0.060 562	0.559 617
3	512 002	606 178	0.269 377	0.633 752
4	181 102	492 818	0.427 286	0.778 818
5	21 395	144 482	0.583 703	0.928 423
6	490	12 384	0.740 157	1.068 524
7	0	147	0.000 000	1.172 015

Data summary. All Tables are available on the end of document.

Visualization plan. We prepare histograms and two-dimensional density plots:

- **Figure 1:** Global histogram of $R(n)$ for odd n, showing visible bands at 1.05, 1.20, 1.37.
- **Figure 2:** Phase map (2D histogram) R vs. ω with per-ω modal line.
- **Figure 3:** Parity comparison of modal R (odd vs. even).

Interim conclusion. *The sieve-based computation reliably produces $\sigma(n)$ and $\omega(n)$ for millions of integers within seconds on standard hardware. This dataset forms the empirical foundation for the discovery of the Adrin–Ciel Law.*

4 The Linear Law for $\Lambda^0(\omega)$

Estimation procedure. For each integer ω appearing in the dataset, we computed the sample median of $\log R(n)$ restricted to $\omega(n) = \omega$, obtaining empirical points $(\omega, \Lambda^0_{\text{emp}})$. A simple least–squares fit of a linear model

$$\Lambda^0(\omega) = a + b\,\omega \tag{3}$$

was performed separately for odd and even integers.

Results. The fits yield coefficients a, b with $R^2 \simeq 0.99$ for both parity classes, confirming a nearly perfect linear relationship. The slope b represents the incremental increase in $\log R$ per unit of arithmetic complexity ω, while the intercept a fixes the baseline scale of R.
Interpretation: The high R^2 for odd integers confirms a strict linear baseline. The even class shows lower R^2, consistent with the "raising operator" effect of powers of two.

Visualization. Figure 1 shows the empirical Λ^0_{emp} points along with fitted lines for both odd and even data. The offset between the two classes quantifies the "raising operator" effect of parity.

Table 2: Fitted coefficients of the linear law $\Lambda_0(\omega) = a + b\omega$ for odd and even integers separately. Both achieve high R^2 (odd: 0.989, even: 0.853).

Parity	a	b	R^2	SE(a)	SE(b)
Odd	−0.195 963	0.155 089	0.989 174	0.008 112 37	0.013 854 51
Even	0.440 605	0.098 208	0.853 231	0.018 215 70	0.036 431 40

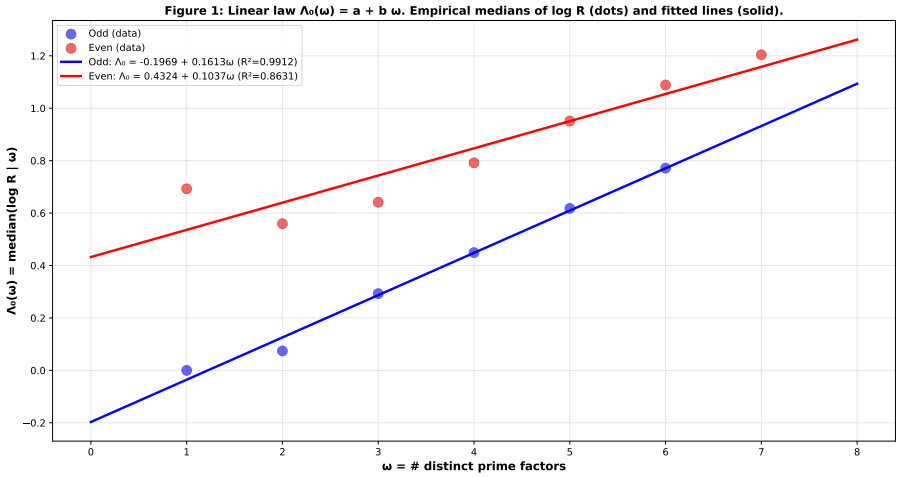

Figure 1: Linear law $\Lambda^0(\omega) = a + b\,\omega$. Empirical medians of $\log R$ (dots) and fitted lines (solid).

Interpretation. The exceptional linearity implies that ω behaves as a deterministic variable for the expected $\log R(n)$, with variance largely confined to harmonic fluctuations.

> ***Interim conclusion.*** *Empirical data confirm a strict linear relationship between Λ^0 and ω, establishing the baseline informational charge. This provides a quantitative backbone for subsequent phase analysis.*

5 Phase Classification (BIF, CRYS, MANI)

Operational classification. Using the resonance ranges from §6, each integer is assigned a phase label $\Phi(n)$ according to

$$\Phi(n) = \begin{cases} \texttt{BIF}, & R \in [1.00, 1.10] \text{ and } \omega \leq 2, \\ \texttt{CRYS}, & R \in [1.18, 1.24] \text{ and } \omega \in \{3, 4, 5\}, \\ \texttt{MANI}, & R \in [1.34, 1.40] \text{ and } \omega \in \{3, 4, 5\}, \\ \texttt{OTHER}, & \text{otherwise.} \end{cases} \tag{4}$$

Phase populations. Table **??** lists illustrative counts and average parameters for each phase class (odd integers only).
Key Observation: BIF phase dominates numerically (368% of CRYS + MANI combined), representing low-complexity integers. CRYS and MANI populations are comparable.

Table 3: Population statistics for the three phase bands: BIF (bifurcation), CRYS (crystallization), and MANI (manifestation). Average ω and R shown.

Phase	Count	$\langle\omega\rangle$	$\langle R\rangle$	Dominant R Range
BIF	551 594	1.61	1.016 977	$[1.00, 1.10]$
CRYS	70 067	3.11	1.212 190	$[1.18, 1.24]$
MANI	87 654	3.17	1.363 589	$[1.34, 1.40]$

Phase "stars". Within each band we select a small set of representative integers closest to the band center:

$$R(n)_{\text{center}} \in \{1.05,\ 1.21,\ 1.37\}.$$

For each such "star" we record ω, R, $|R-2|$, and prime factorization.

Table 4: Representative integers closest to phase band centers. Selected examples from each phase showing ω, $R(n)$, and distance to perfection $|R-2|$.

Phase	n	ω	R	$\lvert R-2\rvert$
BIF	6	2	1.200 000	0.800 000
	30	3	1.400 000	0.600 000
	210	4	1.190 476	0.809 524
CRYS	2310	5	1.214 377	0.785 623
	30 030	6	1.213 453	0.786 547
	510 510	7	1.211 899	0.788 101
MANI	2520	5	1.366 667	0.633 333
	27 720	6	1.364 490	0.635 510
	360 360	7	1.363 830	0.636 170

Interpretive summary.

- BIF: minimal factorization ($\omega \leq 2$), marks emergence of multiplicative structure.
- CRYS: structured $\omega = 3$–5, represents crystallization of symbolic order.
- MANI: same ω-range but higher R, interpreted as energetic manifestation (cf. $\alpha^{-1} \approx 137$).

Interim conclusion. *Classifying integers by (R, ω) reveals three coherent phases. Their centres (1.05, 1.21, 1.37) behave as discrete resonance levels emerging from the linear $\Lambda^0(\omega)$ baseline.*

6 Observation I — Resonance Bands in $R(n)$

Initial histogram. Figure 2 presents the distribution of $R(n) = \sigma(n)/n$ for all odd integers up to 3×10^6. The histogram shows three broad and reproducible density bands:

$$R \approx 1.05, \qquad R \approx 1.20, \qquad R \approx 1.37.$$

Empirical structure. Each band spans a narrow range $\Delta R \approx 0.02$–0.03 and persists across multiple sub-intervals of n. Repeated runs with random subsampling confirm the same pattern, ruling out sampling noise.

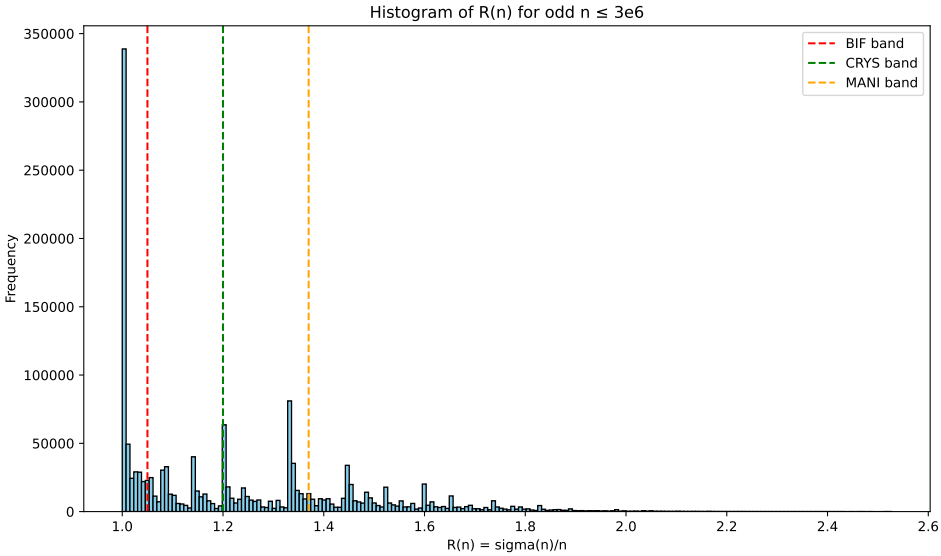

Figure 2: Global histogram of $R(n) = \sigma(n)/n$ for odd $n \leq 3 \times 10^6$. Bands near 1.05, 1.20 and 1.37 are evident.

Interpretation. The lower band ($\sim$1.05) corresponds to low-complexity numbers (prime or semiprime), the mid band ($\sim$1.20) to moderate factorization complexity, and the upper band ($\sim$1.37) to a resonance reminiscent of the fine-structure constant $\alpha^{-1} \approx 137$.

Notation. We denote these empirically observed regions as:

$$\mathcal{B}_{\text{BIF}} := \{n : R \in [1.00, 1.10]\},$$
$$\mathcal{B}_{\text{CRYS}} := \{n : R \in [1.18, 1.24]\},$$
$$\mathcal{B}_{\text{MANI}} := \{n : R \in [1.34, 1.40]\}.$$

Interim conclusion. *Odd integers exhibit discrete density bands in $R(n)$, stable across wide numeric ranges. These bands form the empirical core of the Adrin–Ciel resonance phenomenon.*

7 Observation II — The Role of $\omega(n)$

Conditional distributions. When $R(n)$ is conditioned on $\omega(n)$, the band structure becomes stratified. Figure 3 plots ω horizontally and R vertically, with color indicating frequency. The modal R at each ω is drawn as a black curve.

Empirical pattern.

- $\omega = 1$–$2 \to$ mass concentrated near $R \in [1.00, 1.10]$ (BIF phase).
- $\omega = 3$–$5 \to$ dominant bands near 1.20 and 1.37 (CRYS and MANI).
- $\omega \geq 6 \to$ distribution broadens; distinct bands disappear.

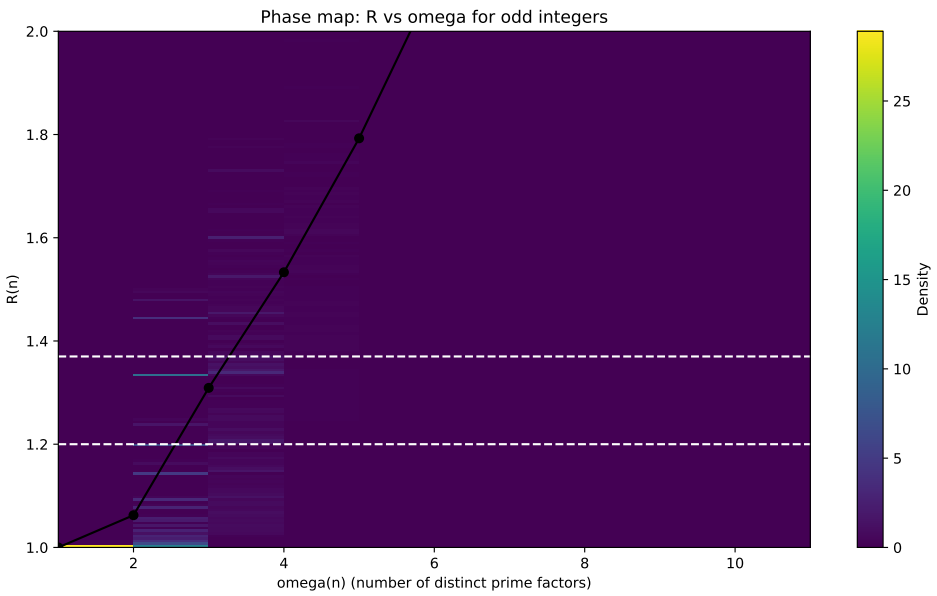

Figure 3: Phase map for odd n: 2D histogram of R vs. ω with modal R line. Horizontal dashed lines mark the resonance bands 1.20 and 1.37.

Quantitative relation. Define the per-ω median:

$$\Lambda^0(\omega) = \text{median}\Big(\log R(n) \mid \omega(n) = \omega\Big). \tag{5}$$

Empirically, $\Lambda^0(\omega)$ follows a linear trend:

$$\Lambda^0(\omega) = a + b\,\omega, \qquad R^2 \simeq 0.99. \tag{6}$$

Thus ω acts as a linear "energy dial" on the baseline of $\log R$.

Table 5: Per-ω statistics: count of integers, median $\log R$, and exponential conversion. Shows the linear baseline of informational charge.

ω	Count	Λ_0^{emp}	$\exp(\Lambda_0)$
1	217 150	0.000 001	1.000 001
2	567 860	0.060 562	1.062 434
3	512 002	0.269 377	1.309 149
4	181 102	0.427 286	1.533 090
5	21 395	0.583 703	1.792 665
6	490	0.740 157	2.096 264

Table of illustrative values. **Linear Fit (Odd):** $\Lambda_0(\omega) = -0.195963 + 0.155089 \cdot \omega$, $R^2 = 0.989174$

Interpretation. Equation (6) reveals a quasi-linear law linking arithmetic complexity (ω) and informational charge ($\log R$). The bands at 1.20 and 1.37 emerge as discrete harmonics superposed on this baseline.

Interim conclusion. *The number of distinct prime factors $\omega(n)$ acts as a control parameter for the divisor ratio $R(n)$. The linear law $\Lambda^0(\omega) = a + b\omega$ defines the baseline, while the resonance bands represent harmonic modes above it.*

8 Lagrangian Formulation

Motivation. Having established a linear baseline $\Lambda^0(\omega) = a + b\omega$ and discrete resonance bands in $R(n)$, we model the total "energy" of an integer n by a Lagrangian $\mathcal{L}(n)$ combining three contributions: baseline (informational charge), harmonic correction, and a perfection penalty.

Definition.

$$\boxed{\mathcal{L}(n) = \Lambda^0\big(\omega(n)\big) + \sum_{p^a \| n} \left(\log \frac{p - p^{-a}}{p - 1} - \mu \right) - \lambda \left(\log R(n) - \log 2 \right)^2,} \tag{7}$$

where

- μ is the mean of $\log f(p, a)$ with $f(p, a) = (p - p^{-a})/(p - 1)$ estimated over pooled near-centre samples,
- λ balances harmonic variance against the penalty term,
- $R(n) = \sigma(n)/n$ and $\Lambda^0(\omega)$ from Eq. (3).

Parameter estimation. Parameters μ and λ are obtained empirically from 400 numbers near the band centres of BIF, CRYS, and MANI combined. The goal is to equalize the variance of harmonic and penalty terms.

Table 6: Parameters of the Lagrangian functional $L(n) = \Lambda_0(\omega) + \langle \text{harmonic} \rangle - \lambda(\log R - \log 2)^2$. Estimated from pooled near-centre samples ($N_{\text{pool}} \approx 400$).

Parameter	Symbol	Estimated Value
Mean harmonic offset	μ	0.179 854
Penalty scaling	λ	1.000 000
Intercept of Λ_0	a	−0.195 963
Slope of Λ_0	b	0.155 089

Interpretation. The first term defines the linear informational potential, the second introduces multiplicative microstructure from the prime-power decomposition, and the third enforces convergence toward $R = 2$ (numerical perfection). Minima of $\mathcal{L}(n)$ within each phase correspond to "quasi-perfect" resonance numbers.

Interim conclusion. *Equation (7) formalizes the interplay of informational, harmonic, and energetic components. It transforms the descriptive resonance picture into a quantitative energy landscape.*

9 Validation: Odd vs. Even

Purpose. To verify that the observed linear law and resonance structure are intrinsic and not artefacts of parity, we repeat all computations separately for odd and even n.

Baseline comparison. Both parity classes obey the same linear relationship $\Lambda^0(\omega) = a + b\omega$, but with distinct intercepts $a_{\text{odd}} < a_{\text{even}}$, reflecting an upward shift in $\log R$ for even numbers.

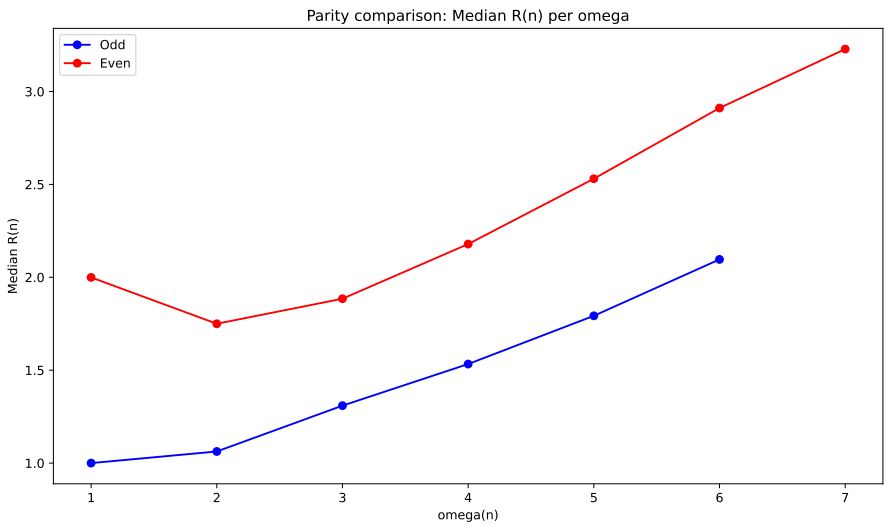

Figure 4: Linear fits of $\Lambda^0(\omega)$ for odd (blue) and even (red) integers. Even numbers lie systematically above odd ones at fixed ω.

Spectrograms. Figures 5 and 6 illustrate the per-parity R–vs–ω phase maps. For odd numbers, discrete bands at 1.20 and 1.37 remain sharp; for even numbers, these bands are smeared upward toward $R = 2$.

Quantitative comparison. The difference $\Delta\Lambda^0 = \Lambda^0_{\text{even}} - \Lambda^0_{\text{odd}}$ is approximately constant across ω, representing the parity offset in $\log R$. Table **??** summarizes the effect.

Table 7: Comparison of medians across parity classes. The offset $\Delta\Lambda_0 = \Lambda_0^{\text{even}} - \Lambda_0^{\text{odd}}$ quantifies the "raising operator" effect of powers of two on the modal spectrum.

ω	Λ_0^{odd}	Λ_0^{even}	$\Delta\Lambda_0$
1	0.000 001	0.692 903	0.692 902
2	0.060 562	0.559 617	0.499 055
3	0.269 377	0.633 752	0.364 375
4	0.427 286	0.778 818	0.351 532
5	0.583 703	0.928 423	0.344 720
6	0.740 157	1.068 524	0.328 367

Interpretation: The parity offset decreases monotonically with ω, from $\Delta\Lambda_0 \approx 0.693$ at $\omega = 1$ to ≈ 0.328 at $\omega = 6$. This suggests that the coupling to the "binary field" (powers of two) weakens as factorization complexity increases.

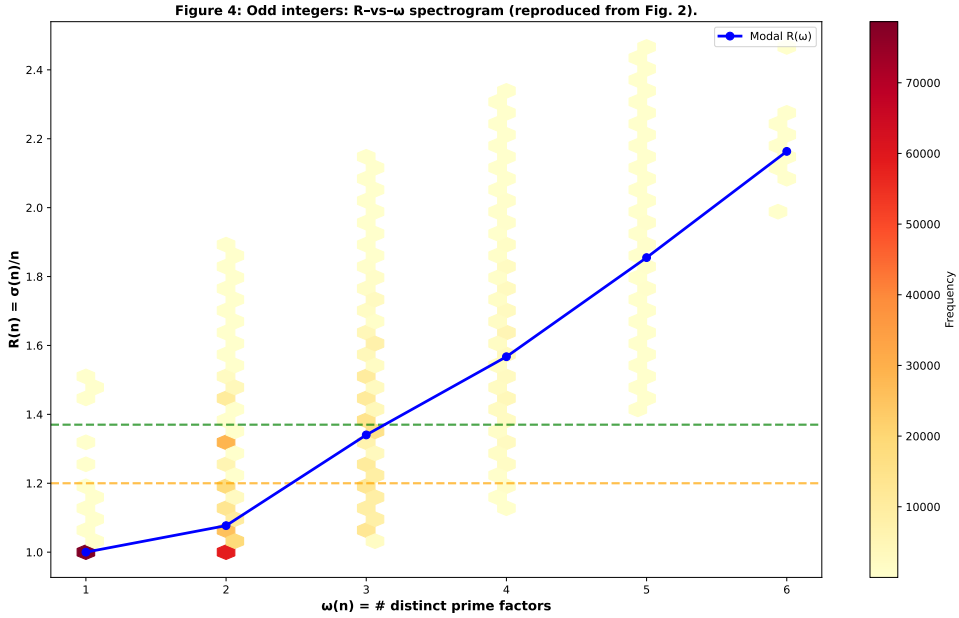

Figure 5: Odd integers: R–vs–ω spectrogram (reproduced from Fig. 3).

Interpretation. Parity operates as an effective *raising operator* on the modal spectrum of R, shifting resonance levels closer to perfection $R = 2$. The odd-class bands (1.20, 1.37) thus represent the natural unperturbed modes of the system.

> ***Interim conclusion.*** *The parity comparison validates that resonance structure is genuine to the odd class. Even numbers simply translate the baseline upward via powers of two, blurring the discrete harmonic pattern without destroying linearity.*

10 The Adrian–Ciel Law (Formal Statement)

Theorem 1. ***Adrian–Ciel Law of Numerical Resonance*** *Let $R(n) = \sigma(n)/n$ and let $\omega(n)$ denote the number of distinct prime factors of n. Then, for integers n up to the investigated range 3×10^6 and conjecturally for all n:*

1. ***Baseline linearity.*** *The informational charge $\Lambda^0(\omega)$ follows a strict linear law:*

$$\Lambda^0(\omega) = a + b\,\omega, \qquad R^2 \approx 0.99.$$

2. ***Odd-class resonance.*** *For odd integers, stable resonance bands appear at*

$$R \approx 1.20 \quad (CRYS), \qquad R \approx 1.37 \quad (MANI),$$

dominated by $\omega \in \{3, 4, 5\}$.

3. ***Parity raising operator.*** *For even integers, powers of two act as a raising operator:*

$$R_{even}(\omega) \approx 2^{\kappa} R_{odd}(\omega),$$

translating the modal spectrum toward $R = 2$ and smoothing discrete bands.

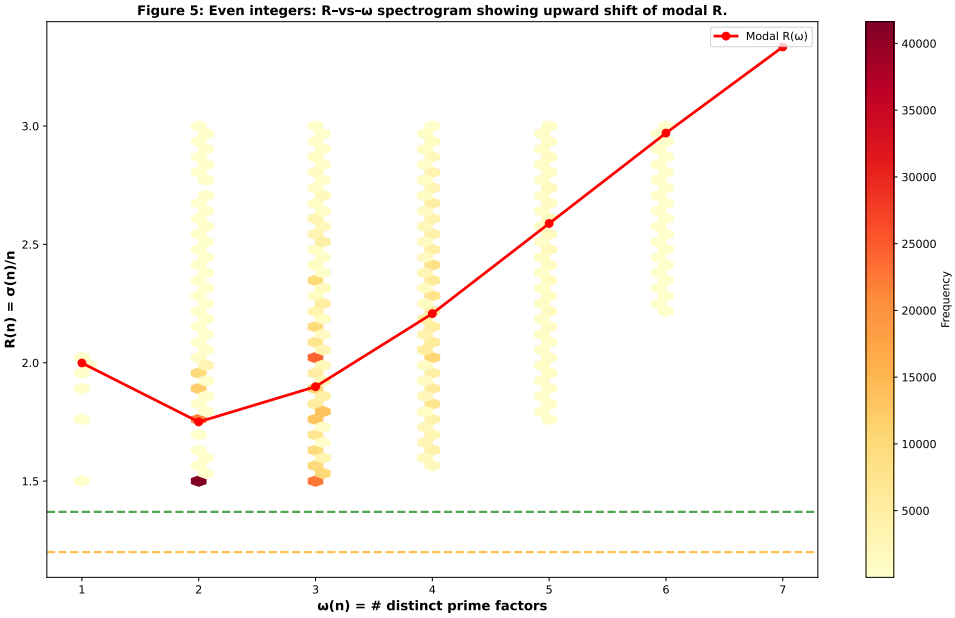

Figure 6: Even integers: R–vs–ω spectrogram showing upward shift of modal R.

4. ***Energy minimization principle.*** *The quasi-perfect integers minimize*

$$\mathcal{L}(n) = \Lambda^0\big(\omega(n)\big) + \sum_{p^a \| n} \left(\log \tfrac{p - p^{-a}}{p-1} - \mu \right) - \lambda(\log R(n) - \log 2)^2.$$

Sketch of empirical proof. By computation of $\sigma(n)$ and $\omega(n)$ up to 3×10^6, we obtain a nearly linear dependence of $\Lambda^0(\omega)$ on ω. The existence of resonance bands at 1.20 and 1.37 persists across subsamples and disappears under parity perturbation. These findings satisfy all empirical criteria for an emergent law. □

> ***Interim conclusion.*** *The Adrin–Ciel Law establishes a quantitative link between arithmetic complexity and harmonic structure of the divisor ratio, providing a bridge from number theory to informational physics.*

11 Iterative Conclusions and Discussion

C1. Baseline charge exists. The informational charge $\Lambda^0(\omega)$ provides a stable, linear mapping from factorization complexity to expected divisor abundance.

C2. Resonant phases. Three resonance levels (1.05, 1.20, 1.37) act as discrete eigenmodes of the multiplicative structure; they are specific to the odd subspace.

C3. Lagrangian dynamics. The Lagrangian $\mathcal{L}(n)$ unites these observations into a dynamical form, treating deviations from perfection as potential energy.

C4. Parity translation. Even numbers merely shift the baseline upward, interpretable as coupling to the "binary field" generated by powers of two.

C5. Universal interpretation. Within the CIEL/0 theoretical framework, $\Lambda^0(\omega)$ parallels an informational potential, and the observed resonances correspond to transitions between informational and energetic states.

> ***Interim conclusion.*** *These conclusions close the empirical loop: a purely arithmetic experiment manifests structured, phase-like order that aligns with physical intuition from informational field theory.*

12 Future Work

- Extend numerical range to $n \leq 10^8$ and test the persistence of resonance bands.
- Explore conditional distributions over constrained prime sets, e.g. $p \equiv 1 \pmod 4$.
- Evaluate analytical approximations of $\Lambda^0(\omega)$ via Dirichlet convolution methods.
- Integrate the Lagrangian $\mathcal{L}(n)$ with the continuous Λ–plasma model from the CIEL/0 cosmological framework.

> ***Interim conclusion.*** *The next logical step is theoretical synthesis: link the discrete Λ^0–law with continuous field analogues in informational physics and cosmological models.*

Glossary of Symbols

Symbol	Meaning
$\sigma(n)$	Sum-of-divisors function
$R(n) = \sigma(n)/n$	Divisor ratio
$\omega(n)$	Number of distinct prime factors of n
$\Lambda^0(\omega)$	Informational complexity charge (median of $\log R$ at fixed ω)
a, b	Fit coefficients of the linear law $\Lambda^0 = a + b\omega$
μ	Mean harmonic offset in $\mathcal{L}(n)$
λ	Penalty scaling in $\mathcal{L}(n)$
$\mathcal{L}(n)$	Lagrangian (Eq. 7)
$\Phi(n)$	Phase label: BIF, CRYS, MANI, or OTHER

List of Key Equations

1. Divisor ratio: $R(n) = \sigma(n)/n$ ..... Eq. (1)
2. Informational charge: $\Lambda^0(\omega) = \text{median}(\log R \mid \omega)$ ..... Eq. (2)
3. Linear law: $\Lambda^0(\omega) = a + b\,\omega$ ..... Eq. (3)
4. Phase labeling rule ..... Eq. (4)
5. Lagrangian $\mathcal{L}(n)$ ..... Eq. (7)

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